Tanmay Bhuskute

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Cincinnati, OH

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production back-end engineering experience in Go, Python, and Java, building distributed systems, REST/GraphQL APIs, and cloud infrastructure on AWS, GCP, and Kubernetes. Strong object-oriented design, problem-solving, and cross-team delivery background.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Go, Python, Java, JavaScript, TypeScript, C/C++, SQL.

Web & Backend: Node.js, Django, Flask, FastAPI, GraphQL, REST APIs, Microservices Architecture.

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch.

Cloud & DevOps: AWS, Google Cloud Platform, Azure, Docker, Kubernetes, ArgoCD, Helm, Terraform, GitLab CI/CD, Jenkins, Linux, Bash.

Data Science & AI: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Gemini, Vertex AI.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and back-end services in a team environment.
- Designed Ansible-based Blue-Green deployment framework, reducing production upgrade downtime by **50%** through reliable rollback and recovery automation.
- Migrated **40+ production environments** to **Kubernetes** using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault, modernizing cloud infrastructure delivery.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PySpark, Delta Lake, FastAPI, PostgreSQL] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.
- Built RESTful API (Python, FastAPI) serving real-time predictions to a React dashboard, integrated with PostgreSQL and Redis caching, containerized with Docker for cloud deployment.